



MIDDLE EAST TECHNICAL UNIVERSITY
NORTHERN CYPRUS CAMPUS

DEPARTMENT
OF
COMPUTER ENGINEERING

CNG 351
Data Management and File Structures
Assignment 4/Quiz Material

FOLLOW UP QUIZ DATE: 19 December 2024, THURSDAY, 17:40 (Cyprus Time)
(8% of actual grade).

PURPOSE

This assignment aims to help you revise SQL in general. **It will NOT be a graded assignment, and you do NOT need to do a submission. However, you will have a follow-up quiz from this assignment.**

This assignment aims to help you revise both Data Definition and Data Manipulation Components of SQL. In the first part, you are given a full database with data, and you need to query the database. This part aims to help you revise especially Select, Aggregation and also Join operations. In the second part, you are given a schema and you need to create the tables, populate them and then query. Therefore, this second part aims to help you revise Create, Insert, Select and Join operations.

IMPORTANT RULES

1. You will need to implement and test all your queries.
2. For implementation, you need to use Oracle Express Edition. In particular, you will need to download and install Oracle 21C Express Edition which is available here: <https://www.oracle.com/database/technologies/xe-downloads.html>. Further instructions are given below.
3. There is NO submission from this assignment. However, a follow-up quiz will be based on the materials and the queries written here.
4. You need to make sure that you complete this assignment before take the quiz which will be on the 19th of December, Thursday at 17:40.

DATABASE CREATION and QUERYING

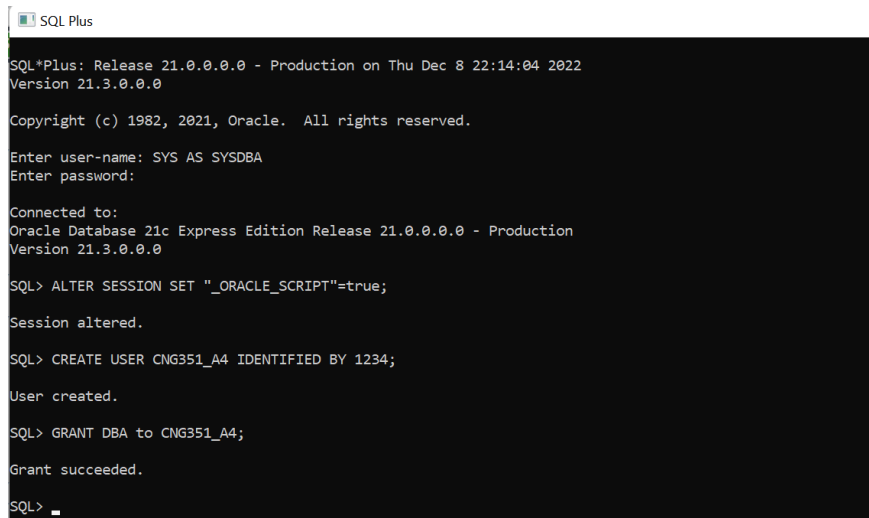
Install and Setup Oracle

In order to continue with the following two use cases, you will need to first install and set up Oracle. Please do the following first:

- First of all, you will need to download and install Oracle 21C from here: <https://www.oracle.com/database/technologies/xe-downloads.html>.
- You will then need to open SQL*Plus. Further information about SQL*Plus is available here: <https://docs.oracle.com/en/database/oracle/oracle-database/21/sqpug/index.html>.
- You will need to execute the following commands to create a user account (replace *user-name* with the one you would like to use as your user name and also replace *password* with your own password) that you can use for this assignment. You need to execute the following commands:

```
sqlplus
SYS AS SYSDBA
ALTER SESSION SET "_ORACLE_SCRIPT"=true;
CREATE USER user-name IDENTIFIED BY password;
GRANT DBA to user-name;
```

Please see the following figure;



```
SQL Plus

SQL*Plus: Release 21.0.0.0.0 - Production on Thu Dec 8 22:14:04 2022
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle. All rights reserved.

Enter user-name: SYS AS SYSDBA
Enter password:

Connected to:
Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> ALTER SESSION SET "_ORACLE_SCRIPT"=true;

Session altered.

SQL> CREATE USER CNG351_A4 IDENTIFIED BY 1234;

User created.

SQL> GRANT DBA to CNG351_A4;

Grant succeeded.

SQL> _
```

Use Case 1 – METU NetRegister

At METU NCC, students must register their devices to our campus network registration system to gain access to the campus network. The university has implemented a comprehensive network registration system called METU NetRegister to manage device registrations and facilitate this process. The main goal of this application is to store information about students, devices, registrations, network engineers, and related network issues and keep track of the registered devices' VPN connection details. Based on the requirements given in the previous assignment, a database was designed and mapped to a logical design as follows. Please note that this is not the direct mapping of the requirements in the previous assignment. You must write some **SQL** queries to address the following queries.

Relations

Student

studentID, name, surname, email, dob, departmentCode

Device

macAddress, model, os, type, owner [FK: Student: studentID]

DeviceSpecification

macAddress [FK: Device: macAddress], specification

Registration

regID, timestamp, status, type, deviceMacAddress [FK: Device: mac_Address], approvedBy [FK: NetworkEngineer: engID]

NetworkEngineer

engID, name, surname, dob

BlockHistory

regID [FK: Registration: regID], startTimestamp, duration, description, blockedBy [FK: NetworkEngineer: engID]

VPNConnectionHistory

regID [FK: Registration: regID], startTimestamp, duration

NetworkIssues

regID [FK: Registration: regID], timestamp, description, comments, resolvedBy [FK: NetworkEngineer: engID]

Domains

Student.department_code = 355, 356, 370, 378

Device.mac_address = A 12-character alphanumeric attribute that is used to identify individual electronic devices on a network.

Device.type = Mobile, Laptop

Registration.status = Approved, Declined, Blocked

Registration.type = Wireless, Wired

Timestamp is stored in the system as “yyyy-mm-dd hh:mm:ss”, e.g., 2024-11-11 18:00:00

BlockHistory.duration = It shows the number of days for blocking

VPNconnectionHistory.duration = It shows the number of hours of connection

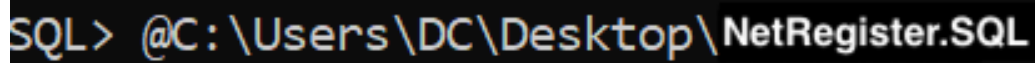
Data Installation

As part of this assignment you are given an SQL file that includes all the commands to create the tables given above, the constraints and also the data. First of all you need to load this file. In order to load it, you will need to execute the following command on SQL*Plus:

```
@path-to-file/NetRegister.sql
```

Please note that *path-to-file* shows the path to the folder which you stored this file. Also, having spaces and Turkish characters in the folder names could cause error

so avoid using spaces and Turkish characters in file or path names. An example screenshot can be seen below.



```
SQL> @C:\Users\DC\Desktop\NetRegister.SQL
```

Queries

1. List the model and operating system (os) of all devices stored in the system.
2. List the Mac addresses of all the approved registrations after 11/11/2024.
3. List the name and surname of the students who own a laptop recorded as a device in the system.
4. List the name and surname of the students whose device registration has declined since the beginning of this year. You can take the beginning of the year as 1/1/2024.
5. A student claims that a network engineer regularly blocks his registered device. We now need to write a query to see if that is correct. We need to identify all the block cases for a student called “Ahmed Zorlu” done by “Mehmet Muhendis”.
6. Our IT manager would like to see a report that shows the total duration of the blocked registrations for wireless and wired registration types. The report will include the total duration for wireless and the total duration for wired blockage.

Use Case 2 – Desire for Velocity

Desire For Velocity is a multiplayer competitive racing game developed by some brilliant indie game developers in METU NCC. It's a real-time racing game with different cars and car classes, races with matchmaking, tournaments, and single-player drift mode. You can find the schema of this database below. Please note that it does not directly map the requirements in the previous assignments. The data to be inserted is given in the file called DesireVelocityData.xlsx.

Schema

You will need to write the necessary commands to create the following tables with all the necessary constraints given below.

Player

player_id, nickname, elo_score, credits

Car

player_id[FK: Player: player_id], car_name, brand, model, class, price, color

Race

race_id, laps, map, weather, tournament_id[FK: Tournament: tournament_id]

Tournament

tournament_id, size, winner[FK: Player, player_id]

Sticker

cosmetic_id, rarity, price, wear float

CarAttachedSticker

player_id[FK: Car: player_id], car_name[FK: Car: name], cosmetic_id[FK: Sticker: cosmetic_id]

CarParticipatesRace

race_id[FK: Race: race_id], player_id[FK: Car: player_id], car_name[FK: Car: name],
duration, position

PlayerNemesis

player1[FK: Player: player_id], player2[FK: Player: player_id], date

Queries

1. List the nickname and elo score for all players in the system.
2. List the players' nicknames that own a car with a class, either 'Electric' or 'Super', and their brand is 'Chevrolet' or 'Tesla'.
3. List the winner's nickname of the tournaments with a size of five.
4. List the sticker details of all the cars for the player called 'SametBaba'. You should report the car name and also the sticker details.
5. List rarity and wear float of all the stickers that belong to the player 'snakeser-dar'.
6. List the names of all players and their car names and their total duration in all races.